

WHAT IS GIT – EASY LEARNING NOTES (BEGINNER FRIENDLY)

1. What is GIT?

GIT is a Version Control System. It is used to track changes in files and folders. GIT does NOT understand any programming language. It works only with files and directories such as Java files, Python files, text files, etc.

2. What does GIT track?

GIT tracks: • What file was changed • Who changed the file • When the change happened • Different versions of the same file

3. How does GIT start tracking files?

GIT starts tracking files only after a repository is created. A repository is created using the command:

git init

4. How to open GIT in a folder?

Steps: 1. Go inside your project folder 2. Right-click inside the folder 3. Click on *Git Bash Here* 4. Git Bash opens (similar to Command Prompt / Terminal)

5. What happens when we run git init?

GIT creates a hidden folder named **.git**. The **.git** folder contains: • Version history • Branch information • Commit details

Important: Without the **.git** folder, the project is NOT a Git repository.

6. Default branch creation

By default, git init creates a branch named **master**. In newer Git versions, the default branch may be **main**.

7. Creating a custom initial branch

If you don't want the default branch name, use:

git init -b main

You can use any branch name, for example:

git init -b demo

8. What happens if git init is executed again?

Git checks the folder and finds that the **.git** folder already exists. It understands that this folder is already a Git repository.

9. What GIT actually does during re-initialization

• It does NOT create a new branch • It ignores the **-b** option • It safely reinitializes the existing repository • No files or commits are deleted

10. Renaming an existing branch

If the current branch is **main** and you want to rename it to **demo**, use:

git branch -m demo

END OF NOTES